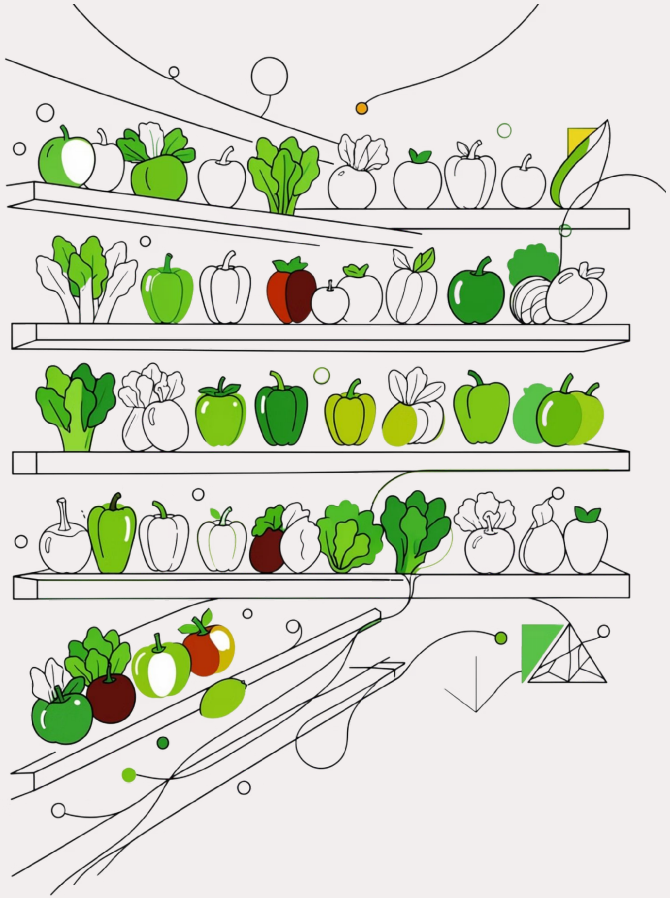
Intelligent Demand Forecasting & Inventory Optimization

A Machine Learning Approach for the Grocery

and

Fast-Moving Consumer Goods Sector

Lubesh Sharma, 22CS30065
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The Dual-Sided Inventory Problem

Grocery retail operates on razor-thin profit margins. The **"Bullwhip Effect"** forces owners into two bad situations:

Overstocking

Ties up capital, increases spoilage (shrinkage), and erodes already thin margins.

Understocking

Causes stockouts, lost revenue, and drives customers to competitors — permanently.

Current State

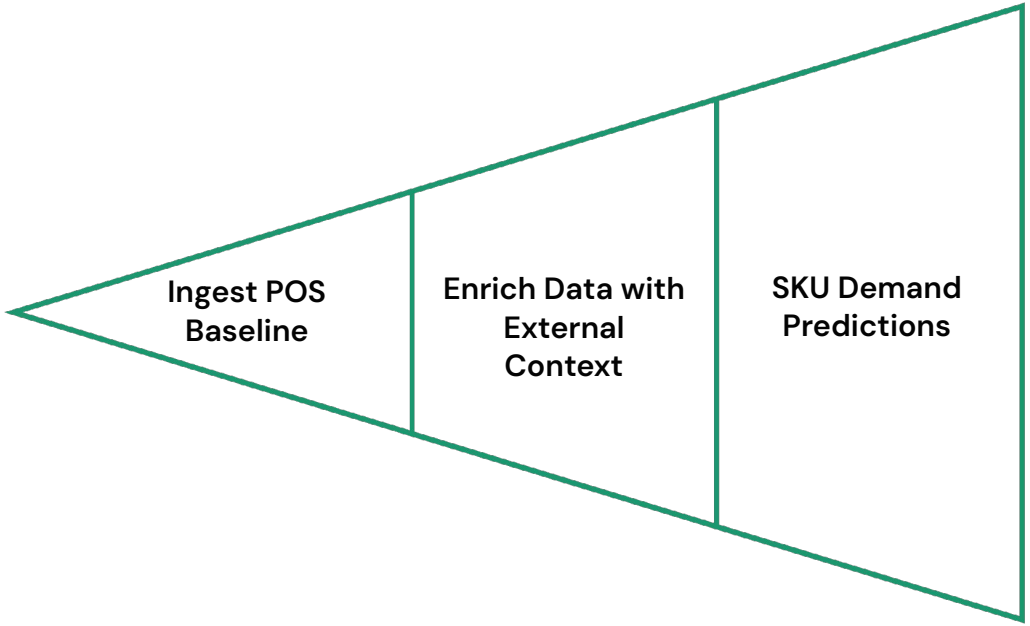
Relying on human intuition or static re-order points — neither scales, neither adapts.

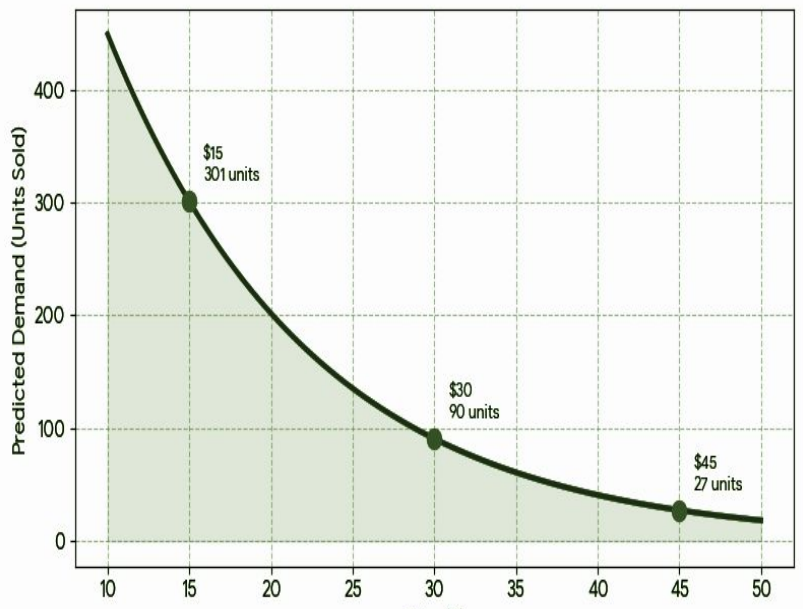


THE SOLUTION

A Context-Aware ML Pipeline

Moving from **'guesswork'** to data-driven predictive insights — at the SKU level.





Target Audience & Use Case



Primary Targets

Independent Supermarkets & digitized **'Kirana'** shops.

Secondary Target

Regional retail chain managers

The Use Case

Wednesday morning ordering. The dashboard detects an upcoming festival + hot weather + payday, and outputs exact order quantities for **500+ items** — automatically.

Datasets & Data Engineering

Primary Dataset 1

[Indian Store](#) Data — **100K Rows** of transactional history.

Primary Dataset 2

[Kolkata Shops Sales](#) — **Grocery Specific** sales records.

Customer ID	Customer Name	Last Name	Date of Birth	Sales	Year	Outlet Type
100000 unique values	Michael 2% David 2% Other (96092) 96%	Smith 2% Johnson 2% Other (96169) 96%	1950-01-01 2018-06-13	101 50k	2019 2023	Medium 34% Large 33% Other (32991) 33%
CUST000001	Curtis	Krause	1973-11-29	3344.69	2023	Large
CUST000002	Jeremy	Foster	1987-08-30	37094.41	2022	Medium
CUST000003	Ariel	Mccann	1980-12-15	35501.88	2023	Large
CUST000004	Colleen	Maynard	1989-05-23	8969.33	2021	Medium
CUST000005	Gina	Holden	1957-12-31	35530.44	2021	Small
CUST000006	Danny	Stevenson	1967-02-25	420.76	2019	Large
CUST000007	Stephanie	Mccoy	2003-10-19	31588.87	2022	Medium
CUST000008	Henry	Cruz	2013-05-22	4419.87	2021	Small
CUST000009	Kristen	Lutz	2005-07-08	41494.09	2023	Large
CUST000010	Gregory	Cuevas	1981-04-10	39717.64	2022	Medium

Programmatic Context Injection

🗓️ Holidays

Python `holidays` library for festival & public holiday flags.

🌦️ Weather

OpenWeatherMap Historical API for temperature & precipitation signals.

Intelligent Modeling Under the Hood

Feature Engineering

Transforming raw data into **Temporal** (Pay-cycles), **Lag** (Historical), and **Contextual** (Weather/Festivals) insights.

The Benchmark Model

ARIMA — Baseline statistical time-series model for performance comparison.

The Core Engine

XGBoost — Excels at tabular data and complex, non-linear relationships.

Stretch Goal

LSTM Neural Networks for sequential mapping of long-range demand patterns.



Targeted Success & Execution Plan

Defining Success (KPIs)

Technical

RMSE & MAPE — forecast accuracy benchmarks.

Business

Reduced holding costs, lowered spoilage, prevented stockouts.

10-Week Execution Timeline

1 **Weeks 1–2**

Data Acquisition, Cleaning, & EDA.

2 **Weeks 3–4**

Feature Engineering & Baseline Models.

3 **Weeks 5–9**

Core XGBoost Modeling / LSTM & KPI Simulations.

4 **Week 10**

Final visualizations and report preparation.

Adapting to Real-World Complexities

Varying Digital Maturity

01

No Data (New Shop)

Lookalike Modeling from comparable stores.

02

Just Digitized

Rapid Baseline Adjustment using early transaction data.

03

Established Shop

Full Multivariate XGBoost model with all context signals.

Key Data Challenges & Mitigations

Zero-Inflated Data

Mitigated via **weekly aggregation** to smooth sparse daily records.

Censored Demand (Stockouts)

Data imputation to recover and estimate lost sales signals.

Concept Drift

Rolling validation windows to keep models current with shifting patterns.

